

From (1) and (2)

$$e^z = e^{z+2n\pi i}$$

Thus  $2\pi i$  is period of  $e^z$ .

### 1.12. Hyperbolic Functions.

We know trigonometric functions like  $\sin \theta$ ,  $\cos \theta$  etc. are connected with a circle. Similarly, there are functions of  $e^x$ , which are connected with a hyperbola and these are known as **Hyperbolic Functions**. These functions have a great similarity to trigonometric functions and are defined as under.

$$(i) \sinh x = \frac{e^x - e^{-x}}{2} \quad (v) \operatorname{sech} x = \frac{1}{\cosh x} = \frac{2}{e^x + e^{-x}}$$

$$(ii) \cosh x = \frac{e^x + e^{-x}}{2} \quad (vi) \operatorname{cosech} x = \frac{1}{\sinh x} = \frac{2}{e^x - e^{-x}}$$

$$(iii) \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

$$(iv) \coth x = \frac{\cosh x}{\sinh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}$$

From (i) and (ii), we have

$$e^x = \cosh x + \sinh x$$

and

$$e^{-x} = \cosh x - \sinh x$$

The following table gives names and nomenclature of these hyperbolic functions.

| Name                    | Abbreviation              | Pronunciation |
|-------------------------|---------------------------|---------------|
| Hyperbolic sin of $x$   | $\sinh x$                 | shin $x$      |
| Hyperbolic cos of $x$   | $\cosh x$                 | cosh $x$      |
| Hyperbolic tan of $x$   | $\tanh x$                 | than $x$      |
| Hyperbolic sec of $x$   | $\operatorname{sech} x$   | shec $x$      |
| Hyperbolic cosec of $x$ | $\operatorname{cosech} x$ | coshec $x$    |
| Hyperbolic cot of $x$   | $\coth x$                 | coth $x$      |

### 1.13 Relation Between Hyperbolic and Circular Functions

$$\begin{aligned} \cos ix &= \frac{1}{2} (e^{i(ix)} + e^{-i(ix)}) \\ &= \frac{1}{2} (e^{-x} + e^x) = \frac{1}{2} (e^x + e^{-x}) = \cosh x. \end{aligned}$$

$$\therefore \cos ix = \cosh x$$

$$\text{Similarly, } \sin ix = i \sinh x,$$

and

$$\tan ix = i \tanh x.$$

Also,

$$\cosh ix = \frac{1}{2} [e^{ix} + e^{-ix}] = \cos x$$

$$\sinh ix = \frac{1}{2} [e^{ix} - e^{-ix}] = i \sin x$$

$$\tanh ix = \frac{\sinh ix}{\cosh ix} = i \tan x \text{ and so on.}$$

**1.14 Relations Between Hyperbolic Functions.**

1.  $\cosh^2 x - \sinh^2 x = 1$
2.  $1 - \tanh^2 x = \operatorname{sech}^2 x$
3.  $\coth^2 x - 1 = \operatorname{cosech}^2 x$
4.  $\sinh (x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y$
5.  $\cosh (x \pm y) = \cosh x \cosh y \pm \sinh x \sinh y$
6.  $\tanh (x \pm y) = \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}$
7.  $\cosh 2x = 2 \cosh^2 x - 1 = 1 + 2 \sinh^2 x$
8.  $\sinh 2x = 2 \sinh x \cosh x$
9.  $\tanh 2x = \frac{2 \tanh x}{1 + \tanh^2 x}$
10.  $\sinh x + \sinh y = 2 \sinh \frac{x+y}{2} \cosh \frac{x-y}{2}$
11.  $\sinh x - \sinh y = 2 \cosh \frac{x+y}{2} \sinh \frac{x-y}{2}$
12.  $\cosh x + \cosh y = 2 \cosh \frac{x+y}{2} \cosh \frac{x-y}{2}$
13.  $\cosh x - \cosh y = 2 \sinh \frac{x+y}{2} \sinh \frac{x-y}{2}$

Proofs of some of the above results are given below.

(1) We know  $\cosh x = \frac{e^x + e^{-x}}{2}$

and  $\sinh x = \frac{e^x - e^{-x}}{2}$

$$\therefore \cosh^2 x - \sinh^2 x = \left( \frac{e^x + e^{-x}}{2} \right)^2 - \left( \frac{e^x - e^{-x}}{2} \right)^2$$

$$= \frac{1}{4} \cdot 4 \cdot e^0 \cdot e^{-0} = 1$$

or  $\cosh^2 x - \sinh^2 x = 1.$

**Aliter.**  $\cos^2 x + \sin^2 x = 1$

Changing  $x$  to  $ix$ , we have

$$(\cos ix)^2 + (\sin ix)^2 = 1$$

or  $\cosh^2 x + (i \sinh x)^2 = 1$

$$\cosh^2 x - \sinh^2 x = 1.$$

(5)  $\cosh (x+y) = \frac{1}{2} \{ e^{x+y} + e^{-(x+y)} \}$

$$= \frac{1}{2} \{ e^x \cdot e^y + e^{-x} \cdot e^{-y} \}$$

$$= \frac{1}{2} [ (\cosh x + \sinh x)(\cosh y + \sinh y) + (\cosh x - \sinh x)(\cosh y - \sinh y) ]$$

$$= \frac{1}{2} [ 2 \cosh x \cosh y + 2 \sinh x \sinh y ]$$

(On simplification)

$$= \cosh x \cosh y + \sinh x \sinh y$$

$$\begin{aligned}
 (7) \quad \cosh 2x &= \frac{1}{2}(e^{2x} + e^{-2x}) \\
 &= \frac{1}{2}[(e^x + e^{-x})^2 - 2] \\
 &= \frac{1}{2}[4 \cosh^2 x - 2] \\
 &= 2 \cosh^2 x - 1 \\
 &= 2(1 + \sinh^2 x) - 1 \\
 &= 1 + 2 \sinh^2 x.
 \end{aligned}$$

$$\begin{aligned}
 (10) \quad \sinh x + \sinh y &= \frac{1}{i} (\sin ix + \sin iy) \\
 &= \frac{1}{i} \left( 2 \sin \frac{ix+iy}{2} \cos \frac{ix-iy}{2} \right) \\
 &= \frac{2i}{i} \sinh \left( \frac{x+y}{2} \right) \cosh \left( \frac{x-y}{2} \right) \\
 &\quad [\because \sin ix = i \sinh x] \\
 &= 2 \sinh \left( \frac{x+y}{2} \right) \cosh \left( \frac{x-y}{2} \right).
 \end{aligned}$$

Similarly we can prove other formulae.

### 1.15. Series Expansions of $\sinh x$ and $\cosh x$

$$\begin{aligned}
 \text{We know} \quad \sinh x &= \frac{e^x - e^{-x}}{2} \\
 &= \frac{1}{2} \left[ \left( 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) \right. \\
 &\quad \left. - \left( 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \right] \\
 &= \frac{1}{2} \left[ 2 \left( x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots \right) \right]
 \end{aligned}$$

$$\therefore \sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

$$\begin{aligned}
 \text{Now} \quad \cosh x &= \frac{1}{2}(e^x + e^{-x}) \\
 &= \frac{1}{2} \left[ \left( 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \right) \right. \\
 &\quad \left. + \left( 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right) \right] \\
 &= \frac{1}{2} \left[ 2 \left( 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots \right) \right]
 \end{aligned}$$

$$\therefore \cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$$

These expansions could easily be obtained by replacing  $x$  by  $ix$  in expansion of  $\sin x$  and  $\cos x$  respectively.

**1.16. Periodicity of Hyperbolic Functions**

Let  $n$  be an integer,

$$\sinh(\theta + 2n\pi i) = \frac{e^{(0+2n\pi i)} - e^{-(0+2n\pi i)}}{2}$$

$$= \frac{e^0 - e^{-0}}{2} = \sinh \theta$$

$$\left[ \begin{array}{l} \because e^{2n\pi i} = \cos 2n\pi + i \sin 2n\pi = 1 \\ e^{-2n\pi i} = \cos 2n\pi - i \sin 2n\pi = 1 \end{array} \right]$$

**Thus period of  $\sinh \theta$  is  $2\pi i$ .**

Similarly it can be shown that period of  $\cosh \theta$  is also  $2\pi i$ .

Also  $\sinh(\theta + \pi i) = \frac{e^{(0+\pi i)} - e^{-(0+\pi i)}}{2}$

$$= -\frac{(e^0 - e^{-0})}{2}$$

$$\left[ \begin{array}{l} \because e^{\pi i} = \cos \pi + i \sin \pi = -1 \\ e^{-\pi i} = \cos \pi - i \sin \pi = -1 \end{array} \right]$$

$$= -\sinh \theta$$

$$\cosh(\theta + \pi i) = \frac{e^{(\theta+\pi i)} + e^{-\theta+\pi i}}{2}$$

$$= \frac{-e^\theta - e^{-\theta}}{2}$$

$$= -\frac{(e^\theta + e^{-\theta})}{2} = -\cosh \theta$$

$$\begin{aligned} \therefore \tanh(\theta + \pi i) &= \frac{\sinh(\theta + \pi i)}{\cosh(\theta + \pi i)} \\ &= \frac{-\sinh \theta}{-\cosh \theta} = \tanh \theta \end{aligned}$$

Similarly  $\coth(\theta + \pi i) = \coth \theta$

**Thus the period of  $\tanh \theta$  and  $\coth \theta$  is  $\pi i$ .**

**Example 1.** Separate into real and imaginary parts.

(iv)  $\sin(x+iy)$       (ii)  $\tan(x+iy)$       (iii)  $\sec(x+iy)$

(iv)  $\sinh(x+iy)$       (v)  $e^{(x-iy)^2}$

**Sol.** (i)  $\sin(x+iy) = \sin x \cos iy + \cos x \sin iy$   
 $= \sin x \cosh y + i \cos x \sinh y$

(ii)  $\tan(x+iy) = \frac{\sin(x+iy)}{\cos(x+iy)}$

Multiplying numerator and denominator on R.H.S. by  $2 \cos(x-iy)$ , we have

$$\tan(x+iy) = \frac{2 \sin(x+iy) \cos(x-iy)}{2 \cos(x+iy) \cos(x-iy)}$$

$$\begin{aligned}
 &= \frac{\sin 2x + \sin 2iy}{\cos 2x + \cos 2iy} \\
 &= \frac{\sin 2x + i \sinh 2y}{\cos 2x + \cosh 2y} \\
 &= \frac{\sin 2x}{\cos 2x + \cosh 2y} + i \frac{\sinh 2y}{\cos 2x + \cosh 2y}
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad \sec(x+iy) &= \frac{1}{\cos(x+iy)} \\
 &= \frac{2 \cos(x-iy)}{2 \cos(x+y) \cos(x-iy)}
 \end{aligned}$$

[Multiplying  $N^r$  and  $D^r$  by  $2 \cos(x-iy)$ ]

$$\begin{aligned}
 &= \frac{2(\cos x \cosh y + i \sin x \sinh y)}{\cos 2x + \cosh 2y} \\
 &= \frac{2 \cos x \cosh y}{\cos 2x + \cosh 2y} + i \frac{2 \sin x \sinh y}{\cos 2x + \cosh 2y}
 \end{aligned}$$

$$\begin{aligned}
 (iv) \quad \sinh(x+iy) &= \frac{1}{i} \sin i(x+iy) \\
 &= \frac{1}{i} \sin(ix-y) = \frac{1}{i} [\sin ix \cos y - \cos ix \sin y] \\
 &= \frac{1}{i} [i \sinh x \cos y - \cosh x \sin y] \\
 &= \sinh x \cos y + i \cosh x \sin y \\
 &\quad \left[ \because -\frac{1}{i} = i \right]
 \end{aligned}$$

$$\begin{aligned}
 (v) \quad e^{(x-iy)^2} &= e^{x^2-y^2-2ixy} = e^{x^2-y^2} \cdot e^{-2ixy} \\
 &= e^{x^2-y^2} (\cos 2xy - i \sin 2xy)
 \end{aligned}$$

**Example 3.** If  $\sin(\alpha+i\beta)=x+iy$ , prove that

$$(i) \quad \frac{x^2}{\cosh^2 \beta} + \frac{y^2}{\sinh^2 \beta} = 1$$

$$(ii) \quad \frac{x^2}{\sin^2 \alpha} - \frac{y^2}{\cos^2 \alpha} = 1.$$

**Sol.** Here  $\sin(\alpha+i\beta)=x+iy$   
or  $\sin \alpha \cosh \beta + i \cos \alpha \sinh \beta = x+iy$

Equating real and imaginary parts, we have

$$x = \sin \alpha \cosh \beta,$$

$$y = \cos \alpha \sinh \beta$$

or  $\frac{x}{\cosh \beta} = \sin \alpha$

and  $\frac{y}{\sinh \beta} = \cos \alpha$

$$\therefore \frac{x^2}{\cosh^2 \beta} + \frac{y^2}{\sinh^2 \beta} = \sin^2 \alpha + \cos^2 \alpha$$

$$\text{or, } \frac{x^2}{\cosh^2 \beta} + \frac{y^2}{\sinh^2 \beta} = 1$$

$$\text{Also } \frac{x}{\sin \alpha} = \cosh \beta$$

$$\text{and } \frac{y}{\cos \alpha} = \sinh \beta$$

$$\text{or } \frac{x^2}{\sin^2 \alpha} - \frac{y^2}{\cos^2 \alpha} = \cosh^2 \beta - \sinh^2 \beta$$

$$\therefore \frac{x^2}{\sin^2 \alpha} - \frac{y^2}{\cos^2 \alpha} = 1.$$

**Example 3.** If  $x+iy = \cosh(u+iv)$ , then prove that

$$(i) \quad \frac{x^2}{\cosh^2 u} + \frac{y^2}{\sinh^2 u} = 1$$

$$(ii) \quad \frac{x^2}{\cos^2 v} - \frac{y^2}{\sin^2 v} = 1.$$

**Sol.** Here  $x+iy = \cosh(u+iv) = \cos(iu-v)$   
 $= \cosh u \cos v + i \sinh u \sin v$

Equating real and imaginary parts, we have

$$x = \cosh u \cos v \quad \dots(1)$$

$$\text{and } y = \sinh u \sin v \quad \dots(2)$$

Eliminating  $v$  from (1) and (2), we have

$$\frac{x^2}{\cosh^2 u} = \cos^2 v \text{ and } \frac{y^2}{\sinh^2 u} = \sin^2 v$$

$$\therefore \frac{x^2}{\cosh^2 u} + \frac{y^2}{\sinh^2 u} = \cos^2 v + \sin^2 v$$

$$\text{or } \frac{x^2}{\cosh^2 u} + \frac{y^2}{\sinh^2 u} = 1.$$

Again eliminating  $u$  from (1) and (2), we have

$$\frac{x^2}{\cos^2 v} - \frac{y^2}{\sin^2 v} = \cosh^2 u - \sinh^2 u$$

$$\therefore \frac{x^2}{\cos^2 v} - \frac{y^2}{\sin^2 v} = 1.$$

**Example 4.** If  $\tan(\theta+i\phi) = \tan \alpha + i \sec \alpha$ , then show that

$$2\theta = n\pi + \frac{\pi}{2} + \alpha$$

$$e^{i\phi} = \pm \cot(\alpha/2)$$

**Sol.** Here  $\tan(\theta+i\phi) = \tan \alpha + i \sec \alpha$

$$\therefore \tan(\theta-i\phi) = \tan \alpha - i \sec \alpha$$

Now  $\tan 2\theta = \tan [(\theta + i\phi) + (\theta - i\phi)]$

$$\begin{aligned} &= \frac{\tan(\theta + i\phi) + \tan(\theta - i\phi)}{1 - \tan(\theta + i\phi)\tan(\theta - i\phi)} \\ &= \frac{(\tan \alpha + i \sec \alpha) + (\tan \alpha - i \sec \alpha)}{1 - (\tan \alpha + i \sec \alpha)(\tan \alpha - i \sec \alpha)} \\ &= \frac{2 \tan \alpha}{1 - (\tan^2 \alpha + \sec^2 \alpha)} = -\frac{2 \tan \alpha}{2 \tan^2 \alpha} \\ &= -\cot \alpha = \tan \left[ \left( n\pi + \frac{\pi}{2} \right) + \alpha \right] \end{aligned}$$

$$\therefore 2\theta = n\pi + \frac{\pi}{2} + \alpha$$

$\tan 2i\phi = \tan [(\theta + i\phi) - (\theta - i\phi)]$

$$\begin{aligned} &= \frac{\tan(\theta + i\phi) - \tan(\theta - i\phi)}{1 + \tan(\theta + i\phi)\tan(\theta - i\phi)} \\ &= \frac{(\tan \alpha + i \sec \alpha) - (\tan \alpha - i \sec \alpha)}{1 + (\tan \alpha + i \sec \alpha)(\tan \alpha - i \sec \alpha)} \\ &= \frac{2i \sec \alpha}{1 + (\tan^2 \alpha + \sec^2 \alpha)} = \frac{2i \sec \alpha}{2 \sec^2 \alpha} = i \cos \alpha \end{aligned}$$

$$\therefore i \tanh 2\phi = i \cos \alpha$$

$$\tanh 2\phi = \cos \alpha$$

$$\text{or } \frac{e^{2\phi} - e^{-2\phi}}{e^{2\phi} + e^{-2\phi}} = \frac{\cos \alpha}{1}$$

Applying componendo-dividendo, we have

$$\frac{e^{2\phi}}{e^{-2\phi}} = \frac{1 + \cos \alpha}{1 - \cos \alpha} = \frac{2 \cos^2 \alpha/2}{2 \sin^2 \alpha/2} = \cot^2 \frac{\alpha}{2}$$

$$\text{or } e^{4\phi} = \cot^2 \alpha/2$$

$$\therefore e^{2\phi} = \pm \cot \alpha/2.$$

**Example 5.** If  $\tan(x + iy) = \sin(u + iv)$ , then prove that

$$\frac{\sin 2x}{\sinh 2y} = \frac{\tan u}{\tanh v}$$

**Sol.** Here  $\tan(x + iy) = \sin(u + iv)$

$$\frac{\sin 2x}{\cos 2x + \cosh 2y} + \frac{i \sinh 2y}{\cos 2x + \cosh 2y} = \sin u \cosh v + i \cos u \sinh v$$

[see example 1 (ii)]

Equating real and imaginary parts on both sides, we have

$$\frac{\sin 2x}{\cos 2x + \cosh 2y} = \sin u \cosh v \quad \dots(1)$$

$$\frac{\sinh 2y}{\cos 2x + \cosh 2y} = \cos u \sinh v \quad \dots(2)$$

and

Dividing (1) by (2), we get

$$\frac{\sin 2x}{\sinh 2y} = \frac{\sin u \cosh v}{\cos u \sinh v}$$

or

$$\frac{\sin 2x}{\sinh 2y} = \frac{\tan u}{\tanh v}.$$

**Example 6.**  $\sin(\theta + i\phi) = \tan \alpha + i \sec \alpha$ , show that  
 $\cos 2\theta \cosh 2\phi = 3$ .

**Sol.** Here  $\sin(\theta + i\phi) = \tan \alpha + i \sec \alpha$

$$\therefore \sin \theta \cosh \phi + i \cos \theta \sinh \phi = \tan \alpha + i \sec \alpha$$

Equating real and imaginary parts, we have

$$\sin \theta \cosh \phi = \tan \alpha \quad \dots(1)$$

$$\text{and} \quad \cos \theta \sinh \phi = \sec \alpha \quad \dots(2)$$

Squaring and subtracting, we have

$$\sin^2 \theta \cosh^2 \phi - \cos^2 \theta \sinh^2 \phi = \tan^2 \alpha - \sec^2 \alpha = -1$$

$$\text{or} \left( \frac{1 - \cos 2\theta}{2} \right) \left( \frac{1 + \cosh 2\phi}{2} \right) - \left( \frac{1 + \cos 2\theta}{2} \right) \left( \frac{\cosh 2\phi - 1}{2} \right) = -1$$

Simplifying, we have

$$\cos 2\theta \cosh 2\phi = 3.$$

### EXERCISE 1 (e)

1. Using Euler's formulae, prove that

$$(i) \sin^2 z + \cos^2 z = 1$$

$$(ii) \cos 2z = \cos^2 z - \sin^2 z$$

$$(iii) \sin 2z = 2 \sin z \cos z$$

2. Show that

$$(i) \{\sin(\alpha + \theta) - e^{i\alpha} \sin \theta\}^n = \sin^n \alpha \cdot e^{-n\theta i}$$

$$(ii) \{\sin(\alpha - \theta) + e^{\pm i\alpha} \sin \theta\}^n \\ = \sin^{n-1} \alpha \{\sin(\alpha - n\theta) + e^{\pm i\alpha} \sin n\theta\}$$

3. Prove that (i)  $\sinh 3\theta = 3 \sinh \theta + 4 \sinh^3 \theta$

$$(ii) \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$(iii) \tanh 3\theta = \frac{3 \tanh \theta + \tanh^3 \theta}{1 + 3 \tanh^2 \theta}.$$

4. Separate into real and imaginary parts

$$(i) \cos(\alpha + i\beta)$$

$$(ii) \cot(\alpha - i\beta)$$

$$(iii) \operatorname{cosec}(\alpha + i\beta)$$

$$(iv) \cosh(\alpha + i\beta)$$

$$(v) \tanh(\alpha + i\beta)$$

$$(vi) e^{\cosh(x+iy)}$$

5. If  $\tan (\alpha+i \beta)=x+i y$ , prove that  

$$x^2+y^2+2 x \cot 2 \alpha=1$$

and

$$x^2+y^2-2 y \coth 2 \beta=-1.$$
6. If  $\tan (\theta+i \phi)=\sin (x+i y)$ , show that  

$$\coth y \sinh 2 \phi=\cot x \sin 2 \theta.$$
7. If  $\sin (\theta+i \phi)=\rho(\cos \alpha+i \sin \alpha)$ , prove that  

$$\rho^2=\frac{1}{2}(\cosh 2 \phi-\cos 2 \theta)$$

and

$$\tan \alpha=\tanh \phi \cot \theta.$$
8. If  $u=\log \tan (\pi / 4+\theta / 2)$ , prove that  
  - (i)  $\tanh \frac{u}{2}=\tan \frac{1}{2} \theta$
  - (ii)  $\theta=-i \log \tan \left(\frac{\pi}{4}+\frac{i u}{2}\right).$
9. If  $\tan (\theta+i \phi)=\cos \alpha+i \sin \alpha$ , prove that  

$$\theta=\frac{n \pi}{2}+\frac{\pi}{4}$$

and

$$\phi=\frac{1}{2} \log \tan \left(\frac{\pi}{4}+\frac{\alpha}{2}\right).$$
10. If  $\tan (x+i y)=\theta+i \phi$ , prove that  

$$\theta^2+\phi^2=-\frac{\cosh ^2 y-\cos ^2 x}{\cosh ^2 y-\sin ^2 x}.$$
11. If  $\cos (\theta+i \phi)=R(\cos \alpha+i \sin \alpha)$ , prove that  

$$\phi=\frac{1}{2} \log \frac{\sin (\theta-\alpha)}{\sin (\theta+\alpha)}.$$
12. If  $\log \sin (\theta+i \phi)=\alpha+i \beta$ , prove that  
  - (i)  $2 \cos 2 \theta=e^{2 \phi}+e^{-2 \phi}-4 e^{2 \alpha}.$
  - (ii)  $\cos (\theta-\beta)=e^{2 \phi} \cos (\theta+\beta).$
13. If  $\tan (x+i \beta)=i$ , where  $\alpha$  and  $\beta$  are real, prove that  $\alpha$  is indeterminate and  $\beta$  is infinite.
14. If  $\sin (\theta+i \phi)=e^{i \alpha}$ , prove that  

$$\cos ^2 \theta=\sinh ^2 \phi=\pm \sin \alpha.$$
15. If  $x=2 \cos \theta \cosh \phi$ ,  $y=2 \sin \theta \sinh \phi$ , prove that  

$$\sec (\theta+i \phi)+\sec (\theta-i \phi)=\frac{4 x}{x^2+y^2}.$$

## 1.17. Inverse Hyperbolic Functions

If  $\sinh z = w$ , then  $z = \sinh^{-1} w$  is known as the inverse hyperbolic sin of  $w$  and pronounced as *shin inverse w*. Similarly other inverse hyperbolic functions are  $\cosh^{-1} w$ ,  $\tanh^{-1} w$  etc.

The inverse hyperbolic functions are many-valued functions but we shall consider only their principal values.

We shall express inverse hyperbolic functions in the logarithmic forms.

$$\text{Let } \sinh^{-1} w = z$$

$$\text{than } w = \sinh z = \frac{e^z - e^{-z}}{2}$$

$$\text{or } 2w = e^z - \frac{1}{e^z}$$

$$\text{or } e^{2z} - 2we^z - 1 = 0$$

This equation is quadratic in  $e^z$ :

$$\therefore e^z = \frac{2w \pm \sqrt{4w^2 + 4}}{2}$$

As  $e^z$  is always positive, we shall take positive sign only

$$\therefore e^z = w + \sqrt{w^2 + 1}$$

$$\text{or } z = \log (w + \sqrt{w^2 + 1})$$

$$\text{or } \sinh^{-1} w = \log (w + \sqrt{w^2 + 1})$$

Similarly we can show,

$$\cosh^{-1} w = \log (w + \sqrt{w^2 - 1})$$

Further, let  $\tanh^{-1} w = z$

$$\therefore w = \tanh z = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

$$\text{or } \frac{w}{1} = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Applying componendo-dividendo, we get

$$\frac{1+w}{1-w} = \frac{2e^z}{2e^{-z}} = e^{2z}$$

$$\therefore 2z = \log \frac{1+w}{1-w}$$

$$\text{or } z = \frac{1}{2} \log \frac{1+w}{1-w}$$

$$\text{or } \tanh^{-1} w = \frac{1}{2} \log \frac{1+w}{1-w}$$

### 1.18. General values of Inverse Hyperbolic Functions

The general values of  $\sinh^{-1} x$ ,  $\cosh^{-1} x$  and  $\tanh^{-1} x$  are denoted by  $\text{Sinh}^{-1} x$ ,  $\text{Cosh}^{-1} x$  and  $\text{Tanh}^{-1} x$  respectively.

These general values are

$$\text{Sinh}^{-1} x = 2n\pi i + (-1)^n \log \{x + \sqrt{x^2 + 1}\}$$

$$\text{Cosh}^{-1} x = 2n\pi i \pm \log \{x + \sqrt{x^2 - 1}\}$$

$$\text{Tanh}^{-1} x = n\pi i + \frac{1}{2} \log \left( \frac{1+x}{1-x} \right).$$

**Example 1.** Separate  $\sin^{-1} (\alpha + i\beta)$  into real and imaginary parts.

**Sol.** Let  $\sin^{-1} (\alpha + i\beta) = x + iy$

$$\therefore \alpha + i\beta = \sin (x + iy) = \sin x \cosh y + i \cos x \sinh y$$

Equating real and imaginary parts, we get

$$\alpha = \sin x \cosh y \quad \dots(i)$$

$$\beta = \cos x \sinh y \quad \dots(ii)$$

$$\therefore \frac{\alpha^2}{\sin^2 x} - \frac{\beta^2}{\cos^2 x} = \cosh^2 y - \sinh^2 y = 1$$

or  $\alpha^2 \cos^2 x - \beta^2 \sin^2 x = \sin^2 x \cos^2 x$

$$\alpha^2 (1 - \sin^2 x) - \beta^2 \sin^2 x = \sin^2 x (1 - \sin^2 x)$$

or  $\sin^4 x - (\alpha^2 + \beta^2 + 1) \sin^2 x + \alpha^2 = 0$

This equation is quadratic in  $\sin^2 x$ ,

$$\therefore \sin^2 x = \frac{1}{2} [(\alpha^2 + \beta^2 + 1) \pm \sqrt{(\alpha^2 + \beta^2 + 1)^2 - 4\alpha^2}]$$

$$x = \sin^{-1} [\pm \sqrt{\frac{1}{2}(\alpha^2 + \beta^2 + 1)} \pm \sqrt{(\alpha^2 + \beta^2 + 1)^2 - 4\alpha^2}]$$

Similarly eliminating  $x$  from (i) and (ii), we have

$$\frac{\alpha^2}{\cosh^2 y} + \frac{\beta^2}{\sinh^2 y} = 1$$

or  $\alpha^2 \sinh^2 y + \beta^2 \cosh^2 y = \cosh^2 y \sinh^2 y$

$$\alpha^2 \sinh^2 y + \beta^2 (1 + \sinh^2 y) = (1 + \sinh^2 y) \sinh^2 y$$

or  $\sinh^4 y - (\alpha^2 + \beta^2 - 1) \sinh^2 y - \beta^2 = 0$

$$\sinh^2 y = \frac{1}{2} [(\alpha^2 + \beta^2 - 1) \pm \sqrt{(\alpha^2 + \beta^2 - 1)^2 + 4\beta^2}]$$

$$\therefore y = \sinh^{-1} [\pm \sqrt{\frac{1}{2}(\alpha^2 + \beta^2 - 1)} \pm \sqrt{(\alpha^2 + \beta^2 - 1)^2 + 4\beta^2}]$$

**Example 2.** Separate into real and imaginary parts

$$\tan^{-1} (\cos \theta + i \sin \theta).$$

**Sol.** Let  $\tan^{-1} (\cos \theta + i \sin \theta) = A + iB$

$$\therefore \cos \theta + i \sin \theta = \tan (A + iB) \quad \dots(1)$$

and  $\cos \theta - i \sin \theta = \tan (A - iB) \quad \dots(2)$

$$\text{Now } \tan 2A = \tan \{(A + iB) + (A - iB)\}$$

$$= \frac{\tan(A + iB) + \tan(A - iB)}{1 - \tan(A + iB) \tan(A - iB)}$$

$$= \frac{(\cos \theta + i \sin \theta) + (\cos \theta - i \sin \theta)}{1 - (\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta)}$$

[from (1) and (2)]

$$= \frac{2 \cos \theta}{1 - (\cos^2 \theta + \sin^2 \theta)}$$

$$= \frac{2 \cos \theta}{0} = \infty$$

$$\left[ \because \tan \frac{\pi}{2} = \infty \right]$$

$$\tan 2A = \tan \left( n\pi + \frac{\pi}{2} \right)$$

$$\therefore A = \frac{n\pi}{2} + \frac{\pi}{4}$$

$$\text{Now } \tan 2iB = \tan \{(A + iB) - (A - iB)\}$$

$$= \frac{\tan(A + iB) - \tan(A - iB)}{1 + \tan(A + iB) \tan(A - iB)}$$

$$= \frac{(\cos \theta + i \sin \theta) - (\cos \theta - i \sin \theta)}{1 + (\cos \theta + i \sin \theta)(\cos \theta - i \sin \theta)}$$

$$= \frac{2i \sin \theta}{1 + (\cos^2 \theta + \sin^2 \theta)} = i \sin \theta$$

$$\therefore i \tanh 2B = i \sin \theta$$

$$\text{or } \tanh 2B = \sin \theta$$

$$\text{or } \frac{e^{2B} - e^{-2B}}{e^{2B} + e^{-2B}} = \frac{\sin \theta}{1}$$

Applying componendo-dividendo,

$$\frac{2e^{2B}}{2e^{-2B}} = \frac{1 + \sin \theta}{1 - \sin \theta}$$

$$\therefore e^{4B} = \frac{1 + \cos \left( \frac{\pi}{2} - \theta \right)}{1 - \cos \left( \frac{\pi}{2} - \theta \right)} = \frac{2 \cos^2 \left( \frac{\pi}{4} - \frac{\theta}{2} \right)}{2 \sin^2 \left( \frac{\pi}{4} - \frac{\theta}{2} \right)}$$

$$= \cot^2 \left( \frac{\pi}{4} - \frac{\theta}{2} \right)$$

$$\therefore e^{2B} = \cot \left( \frac{\pi}{4} - \frac{\theta}{2} \right)$$

$$\therefore B = \frac{1}{2} \log \cot \left( \frac{\pi}{4} - \frac{\theta}{2} \right)$$

$$\therefore \tan^{-1} (\cos \theta + i \sin \theta) = \frac{n\pi}{2} + \frac{\pi}{4} + \frac{i}{2} \log \cot \left( \frac{\pi}{4} - \frac{\theta}{2} \right).$$

**Example 3.** If  $\cos^{-1} (u+iv) = \alpha + i\beta$ , prove that  $\cos^2 \alpha$  and  $\cosh^2 \beta$  are the roots of the equation

$$x^2 - x(1+u^2+v^2) + u^2 = 0.$$

**Sol.** Now  $\cos^{-1} (u+iv) = \alpha + i\beta$

$$u+iv = \cos (\alpha + i\beta)$$

$$= \cos \alpha \cosh \beta - i \sin \alpha \sinh \beta$$

$$u = \cos \alpha \cosh \beta \quad \dots(i)$$

and

$$v = -\sin \alpha \sinh \beta$$

Now

$$\begin{aligned} u^2 + v^2 &= \cos^2 \alpha \cosh^2 \beta + \sin^2 \alpha \sinh^2 \beta \\ &= \cos^2 \alpha \cosh^2 \beta + (1 - \cos^2 \alpha)(\cosh^2 \beta - 1) \\ &= \cos^2 \alpha + \cosh^2 \beta - 1 \end{aligned}$$

or

$$1 + u^2 + v^2 = \cos^2 \alpha + \cosh^2 \beta$$

also

$$u^2 = \cos^2 \alpha \cosh^2 \beta. \quad \text{from (i)}$$

The equation which has roots  $\cos^2 \alpha$  and  $\cosh^2 \beta$  is,

$$x^2 - (\cos^2 \alpha + \cosh^2 \beta)x + \cos^2 \alpha \cosh^2 \beta = 0$$

or

$$x^2 - (1 + u^2 + v^2)x + u^2 = 0.$$

### 1.19. Logarithm of Complex Quantities

If  $z = x + iy$  and  $w = e^z$ , then  $z$  is defined as  $\log_e w$

$$\text{Now } e^{z+2n\pi i} = e^z \cdot e^{2n\pi i}$$

$$= e^z = w.$$

$$[ \because e^{2n\pi i} = 1 ]$$

$$\therefore z + 2n\pi i = \log_e w$$

[By definition]

Thus we see that  $\log_e w$  is a multivalued function. Hence logarithm of a complex quantity is a multivalued function.

The value of  $\log_e w$  corresponding to  $n=0$ , is called **principal value**. The general value of  $\log_e w$  denoted by  $\log_e w$ , and  $\log_e w = 2n\pi i + \log_e w$ , where  $\log_e w$  denotes the principal value.

**Example 1.** Separate  $\log_e (x+iy)$  into real and imaginary parts and hence obtain the real and imaginary parts of  $\log_e (x+iy)$ .

**Sol.** Let  $\log_e (x+iy) = x + i\beta$

$$\therefore x+iy = e^{x+i\beta} = e^x \cdot e^{i\beta}$$

or

$$x+iy = e^x \cdot (\cos \beta + i \sin \beta)$$

Equating real and imaginary parts, we get

$$x = e^x \cdot \cos \beta \quad \dots (1)$$

$$y = e^x \cdot \sin \beta \quad \dots (2)$$

and

Squaring and adding (1) and (2), we have

$$x^2 + y^2 = e^{2x}$$

$$\therefore 2\alpha = \log_e (x^2 + y^2)$$

$$\alpha = \frac{1}{2} \log_e (x^2 + y^2)$$

or

Also from (1) and (2), we have

$$\tan \beta = \frac{y}{x}$$

or

$$\beta = \tan^{-1} \frac{y}{x}$$

$$\therefore \log_e (x + iy) = \frac{1}{2} \log_e (x^2 + y^2) + i \tan^{-1} (y/x).$$

$$\text{Now } \text{Log}_e (x + iy) = 2n\pi i + \log_e (x + iy)$$

$$= \frac{1}{2} \log_e (x^2 + y^2) + i (2n\pi + \tan^{-1} y/x).$$

**Example 2.** Find the principal values of  $\log (-1 + i) - \log (-1 - i)$ .

**Sol.** Here  $\log (-1 + i) - \log (-1 - i)$

Let  $-1 = r \cos \theta$  and  $1 = r \sin \theta$

$$\therefore r = \sqrt{2}, \tan \theta = -1 \text{ or } \theta = \frac{3\pi}{4}$$

$$\therefore -1 + i = r(\cos \theta + i \sin \theta) = re^{i\theta}$$

$$= \sqrt{2} e^{\frac{3\pi i}{4}}$$

$$-1 - i = r(\cos \theta - i \sin \theta) = re^{-i\theta}$$

$$= \sqrt{2} e^{-\frac{3\pi i}{4}}$$

$$\therefore \log (-1 + i) - \log (-1 - i) = \log \frac{(-1 + i)}{(-1 - i)}$$

$$= \log \frac{\sqrt{2} e^{\frac{3\pi i}{4}}}{\sqrt{2} e^{-\frac{3\pi i}{4}}}$$

$$= \log \frac{1}{e^{-\frac{3\pi i}{2}}} = -\log e^{-\frac{3\pi i}{2}}$$

$$= -\log \left( e^{2\pi i - \frac{3\pi i}{2}} \right) = -\log e^{\frac{\pi i}{2}}$$

$$= -\frac{\pi i}{2}, \text{ is the principal value.}$$

**Example 3.** Show that (i)  $\text{Log } i = \left( 2n + \frac{1}{2} \right) \pi i$

(ii)  $\text{Log } (-x) = (2n + 1) \pi i + \log x$ .

**Sol.** (i)  $i = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2} = e^{\frac{\pi i}{2}}$

$$\log i = \log e^{\frac{\pi}{2}i} = \frac{\pi}{2}i$$

Now  $\text{Log } i = 2n\pi i + \log i$   
 $= 2n\pi i + \frac{\pi i}{2}$   
 $= \left(2n + \frac{1}{2}\right)\pi i.$

(ii)  $-x = x(\cos \pi + i \sin \pi) = xe^{i\pi}$

$$\log(-x) = \log x + \log e^{\pi i}$$

$$= \log x + \pi i$$

$\therefore \text{Log}(-x) = 2n\pi i + \log(-x)$   
 $= 2n\pi i + \log x + \pi i$   
 $= (2n+1)\pi i + \log x$

**Example 4.** Prove that  $\tan \left( i \log \frac{a-ib}{a+ib} \right) = \frac{2ab}{a^2-b^2}$

**Sol.** Let  $a = r \cos \theta$ ,  $b = r \sin \theta$

$\therefore r = \sqrt{a^2 + b^2}$ ,  $\tan \theta = \frac{b}{a}$

$\therefore a + ib = r(\cos \theta + i \sin \theta) = re^{i\theta}$   
 $a - ib = r(\cos \theta - i \sin \theta) = re^{-i\theta}$

$$\frac{a-ib}{a+ib} = \frac{re^{-i\theta}}{re^{i\theta}} = e^{-2i\theta}$$

$$\log \frac{a-ib}{a+ib} = \log e^{-2i\theta} = -2i\theta$$

$$i \log \frac{a-ib}{a+ib} = i(-2i\theta) = 2\theta$$

$$= 2 \tan^{-1} \frac{b}{a}$$

$$= \tan^{-1} \frac{b}{a} + \tan^{-1} \frac{b}{a}$$

$\therefore \tan \left[ i \log \frac{a-ib}{a+ib} \right] = \tan \left[ \tan^{-1} \frac{b}{a} + \tan^{-1} \frac{b}{a} \right]$

$$= \tan \left[ \tan^{-1} \frac{\frac{b}{a} + \frac{b}{a}}{1 - \frac{b^2}{a^2}} \right]$$

$$= \tan \left[ \tan^{-1} \frac{2ab}{a^2 - b^2} \right]$$

$$\therefore \tan \left[ i \log \frac{a - ib}{a + ib} \right] = \frac{2ab}{a^2 - b^2}$$

**Example 5.** If  $i^{i^{\dots\infty}} = A + iB$ , prove that

$$\tan \frac{\pi A}{2} = \frac{B}{A}$$

and

$$A^2 + B^2 = e^{-\pi B}$$

**Sol.** Here  $i^{i^{\dots}} = A + iB$

$$\therefore i^{A + iB} = A + iB$$

Taking logs on both sides, we have

$$(A + iB) \log i = \log (A + iB)$$

$$(A + iB) \left[ \frac{1}{2} \log 1 + i \tan^{-1} \frac{1}{0} \right] = \log (A + iB)$$

$$\left[ \because \log (A + iB) = \frac{1}{2} \log (A^2 + B^2) + i \tan^{-1} \frac{B}{A} \right]$$

$$(A + iB) \left( \frac{\pi}{2} i \right) = \frac{1}{2} \log (A^2 + B^2) + i \tan^{-1} \frac{B}{A}$$

Equating real and imaginary parts on both sides, we have

$$\therefore \frac{-B\pi}{2} = \frac{1}{2} \log (A^2 + B^2)$$

and

$$\frac{\pi A}{2} = \tan^{-1} \frac{B}{A}$$

$\therefore$

$$A^2 + B^2 = e^{-\pi B}$$

and

$$\tan \frac{\pi A}{2} = \frac{B}{A}$$

### EXERCISE 1 (f.)

1. Separate into real and imaginary parts.

(i)  $\sin^{-1} (\cos \theta + i \sin \theta)$

(ii)  $\cos^{-1} (\cos \theta + i \sin \theta)$

(iii)  $\tanh^{-1} (x + iy)$ .

2. Prove that  $\tanh^{-1} x = \sinh^{-1} \left( \frac{x}{\sqrt{1 - x^2}} \right)$ .

3. Prove that  $\coth^{-1} \frac{2}{x} = \sinh^{-1} \left( \frac{x}{\sqrt{4 - x^2}} \right)$ .

4. Show that  $\sin^{-1} (ix) = 2n\pi + i \log [\sqrt{1 + x^2} + x]$  and hence show that

$$\sin^{-1} (i) = 2n\pi + i \log (\sqrt{2} + 1).$$